PARISHRAM 2.0

CLASS 12TH 2026

Electrostatic Potential and Capacitance

PHYSICS LECTURE-1

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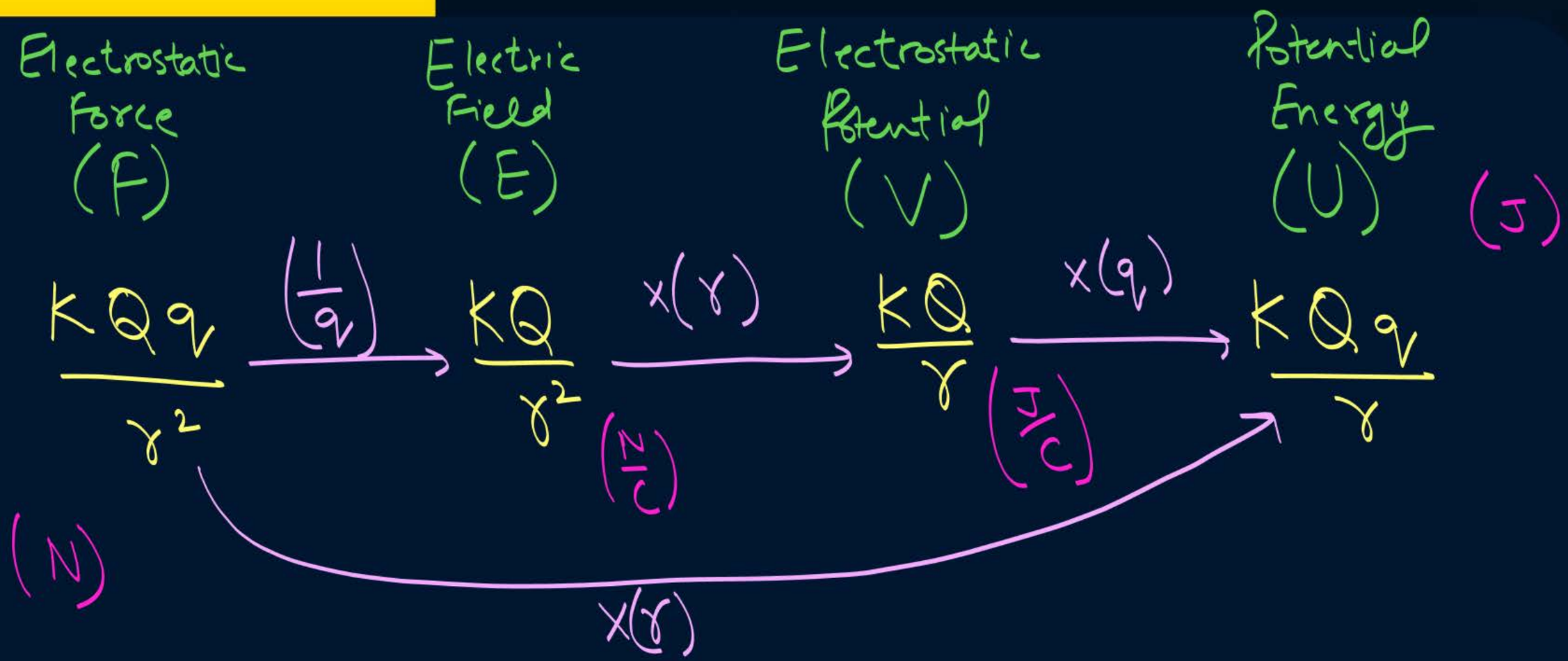

Finally **pehla** chapter
khatam ho gya...


YOU HAVE
POTENTIAL

Topics to be covered:

1. Electrostatic potential
2. Potential difference
3. Electric potential due to
   1. A point charge
   2. A dipole
   3. System of charges

Rough Relations



Introduction

~~$W = F \times d$~~

~~Force~~ ~~Displacement.~~

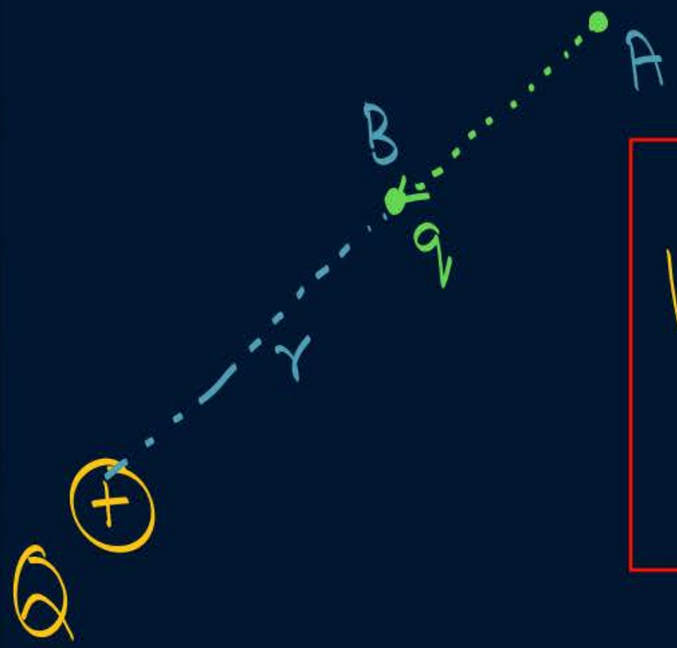
Force = Constant.

$W = \vec{F} \cdot \vec{d}$

Dot Product or Scalar Product.

Force = Variable.

$W = \int \vec{F} \cdot \vec{dr}$



$$W = -\int_{r_A}^{r_B} F_{ext} \cdot dr$$

→ Work done is a Scalar Quantity.

Electrostatic Potential



1 C charge

Definition

Work done to bring a unit **positive** charge from **infinite distance** to a point in the **electric** field is called **electric potential at that point**.

(i) Positively charged body is considered to be at **higher potential** and **negatively** charged body is considered to be at **lower** potential.

(ii) **Electric potential** at a point is a **relative value not** an **absolute value**.

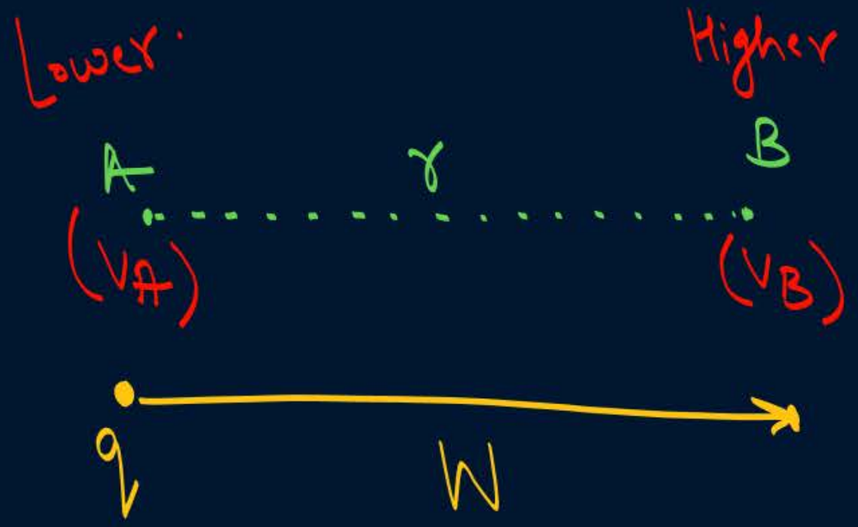
(iii) **Potential** due to a group of charges is the **algebraic sum** of **their** individual potentials.

i.e., $V = V_1 + V_2 + V_3 + _____$.

★ Electrostatic Potential is also a Scalar Quantity.

Potential Difference

Potential difference between **two** points is the **work** done in **moving** a unit **positive** **charge** from **one** point to the **other** in the **electric field**.



$$V = \frac{W}{q}$$

$$\Delta V = V_B - V_A = \frac{W}{q}$$

Potential Difference

Units and Dimensions

- **Unit:** Volt (V).
- **Dimensions:** $[ML^2T^{-3}A^{-1}]$.

Same as that of Electrostatic Potential.

Everyday Example

- A **battery** maintains a potential difference **between** its **terminals**, which **pushes charges** through a **circuit**.
- **1.5 V** battery means **1.5 J** of work is done in moving **1 C** charge between its **terminals**.

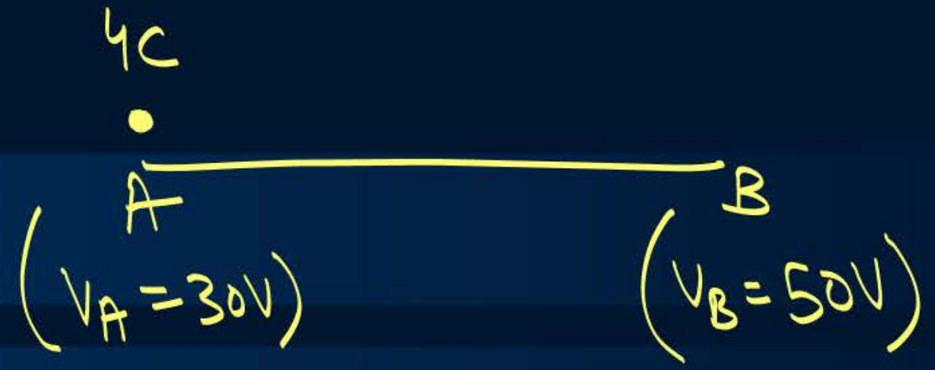


By Work - Energy theorem :-

$$W = E_f - E_i$$

QUESTION

A charge of **4 C** is moved from a point at potential **30 V** to a point at potential **50V**. Find the **work** done.



$$\frac{W}{q} = \Delta V$$



A. 40 J

B. 60 J

C. 80 J

D. 100 J

$$\frac{W}{q} = \Delta V = V_B - V_A = 50 - 30 = 20V$$

$$\Rightarrow W = 4 \times 20 \Rightarrow W = 80\,J$$

QUESTION

A proton is accelerated from rest through a potential difference of 1000 V. Find its kinetic energy gained. (Charge of proton = 1.6×10^{-19} C)

$W = \Delta K.E.$

$\Delta K.E. = q \cdot \Delta V$

$= 1.6 \times 10^{-19} \times 1000$

$= 1.6 \times 10^{-16}$

A 1.6×10^{-16} J

B 1.6×10^{-19} J

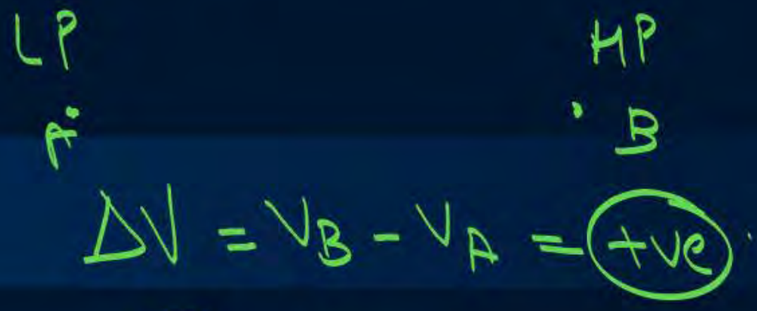
C 1.6×10^{-20} J

D 1.6×10^{-22} J



QUESTION

When a positive q charge is taken from lower potential to a higher potential point, then its potential energy will:

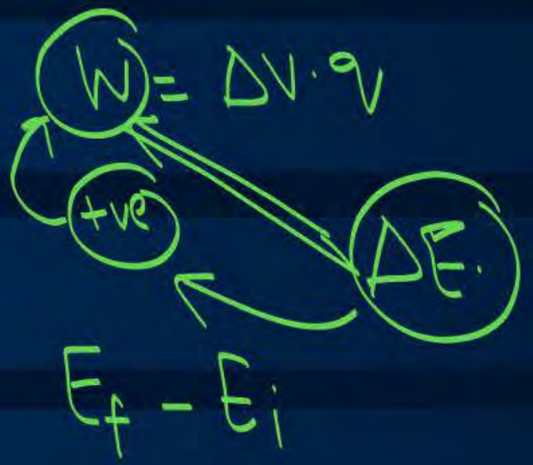


A decrease

B increases

C remain unchanged

D become zero





Electrostatic Potential due to a point charge :—



$$V = W_{\infty P} = E_P - E_\infty \quad \left(\text{By Work-Energy Theorem.}\right)$$

$$V = -\int_{r=\infty}^{r=r} F_{ext}\cdot dr = -\int_{r=\infty}^{r=r} \frac{Kq(1)}{r^2}\,dr$$

$$V = -Kq\int_{r=\infty}^{r=r} \frac{1}{r^2}dr = -Kq\left[\frac{-1}{r}\right]_{r=\infty}^{r=r}$$

$$V = -Kq\left[\frac{-1}{r} + \frac{1}{\infty}\right] \Rightarrow \boxed{V = \frac{Kq}{r}}$$

→ Electrostatic Potential due to a point charge q.

$$\left[K = \frac{1}{4\pi\varepsilon_0} = 9\times10^9\ Nm^2/C^2\right]$$

Potential due to a Point Charge

Example

Calculate potential due to a charge 5×10^{-7} C at a point P located 9 cm away. Also find wok done in bringing an another charge of 5×10^{-8} C from infinity to the point.

$$V = \frac{kq}{r} = \frac{9 \times 10^9 \times 5 \times 10^{-7}}{9 \times 10^{-2}} = 5 \times 10^4 \text{ V}$$

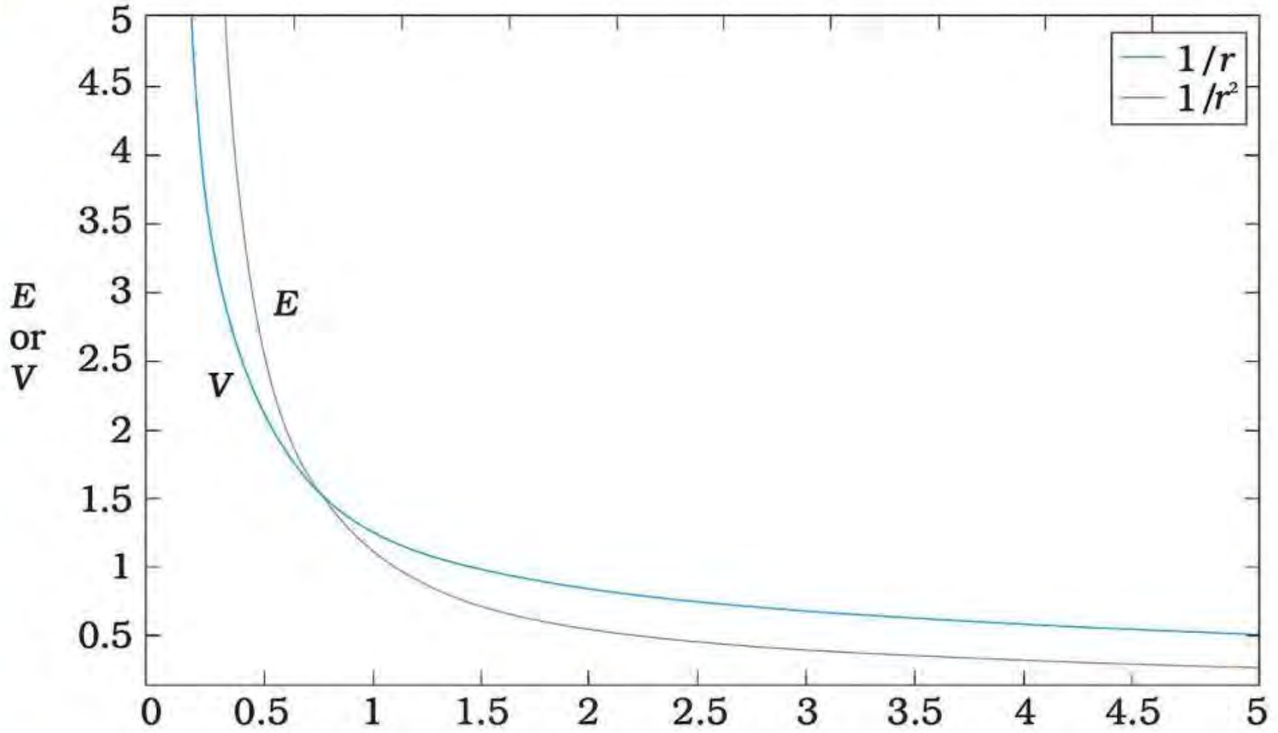
$$5 \times 10^4 \text{ V} = \frac{W}{5 \times 10^{-8} \text{ C}} \Rightarrow W = 5 \times 10^4 \times 5 \times 10^{-8} \text{ J}$$

$$W = 25 \times 10^{-4} \text{ J}$$



Potential due to a Point Charge

Revision



Variation of potential V with *r* [in units if $(Q/4\pi\varepsilon_0)$ m^{-1}) (Blue curve)] and field with *r* [in units if $(Q/4\pi\varepsilon_0)$ m^{-2}) (Black curve)] for a point charge Q.

QUESTION

The potential at a point, due to a positive charge of 100 μC at a distance of 9 m, is;

$$V = \frac{Kq}{r} = \frac{9 \times 10^9 \times 100 \times 10^{-6}}{9}$$

$V = 10^5$ V

A 10^4 V

B 10^5 V

C 10^6 V

D 10^7 V



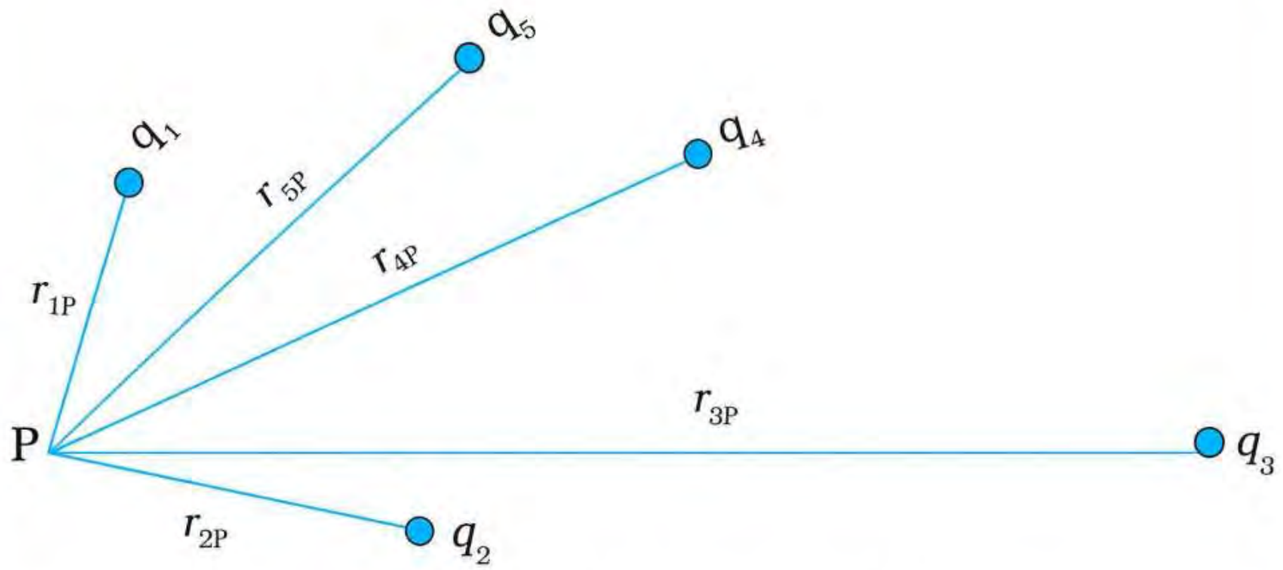
Potential due to System of Charges

ex:



$$V_{net} = V_1 + V_2 + V_3 + V_4$$

$$V_{net} = K\left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \frac{q_4}{r_4}\right]$$



Potential at a point due to a system of charges is the sum of potentials due to individual charges.

Potential due to System of Charges

NCERT

If we have a continuous charge distribution characterized by a charge **density ρ (r)**, we divide it, as before, into small volume elements each of size Δv and carrying a charge $\rho \Delta v$. We then determine the potential due to each volume element and sum (strictly speaking, integrate) over all such contributions, and thus determine the potential due to the entire distribution.

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \quad (r \geq R)$$

We have seen in Chapter 1 that for a uniformly charged spherical shell, the electric field outside the shell is as if the entire charge is concentrated at the centre. Thus, the potential outside the shell is given by

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{R}$$

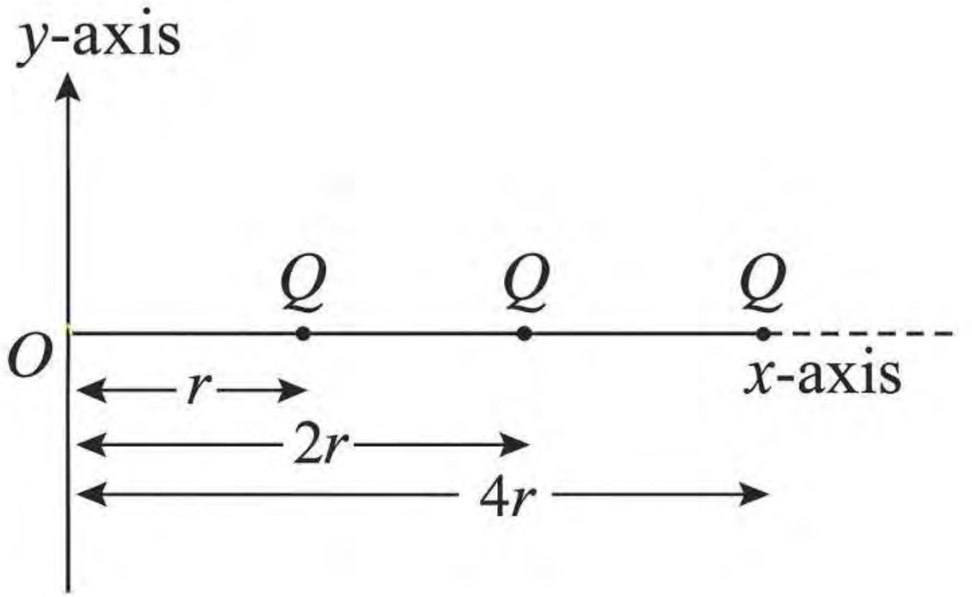
Potential due to System of Charges

$$(V_{net}) = \frac{kQ}{r} + \frac{kQ}{2r} + \frac{kQ}{4r} + \frac{kQ}{8r} + \cdots$$

$$V = \frac{kQ}{r}\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots\right)$$

$$V = \frac{kQ}{r}\left(\frac{1}{1 - \frac{1}{2}}\right)$$

$$\Rightarrow \boxed{V = \frac{2kQ}{r}}$$



Geometric Progression :-

ex: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots\ldots \infty$ (infinite GP).

$SUM = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots\ldots \infty$

$SUM = \left(\frac{a}{1-r}\right)$ where, a = first element of the GP
r = Common ratio of the GP.

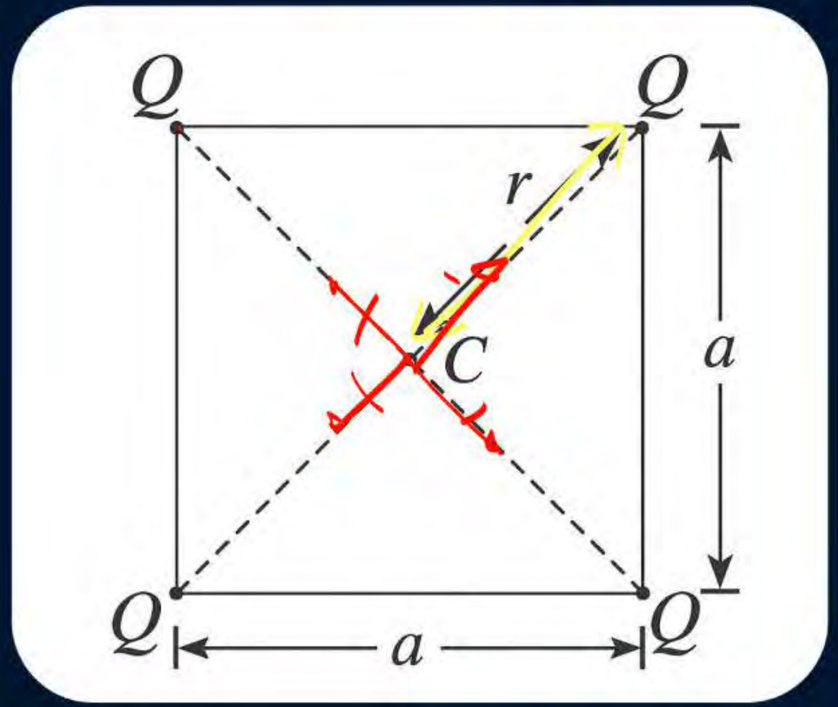
$\Rightarrow SUM = \left(\frac{1}{1-\frac{1}{2}}\right) = \boxed{2}$

Potential due to System of Charges

$$r = \frac{a\sqrt{2}}{2} = \left(\frac{a}{\sqrt{2}}\right)$$

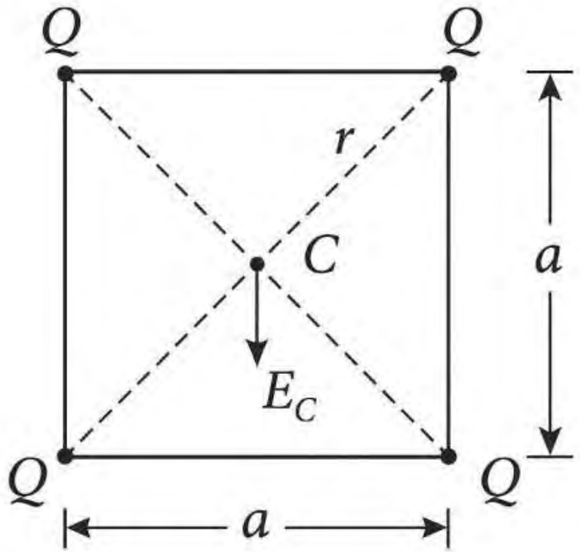
$$V_C = \frac{kQ}{\left(\frac{a}{\sqrt{2}}\right)} \times 4 \Rightarrow \boxed{V_C = \frac{4\sqrt{2}\,kQ}{a}}$$

$$\boxed{E_C = 0}$$



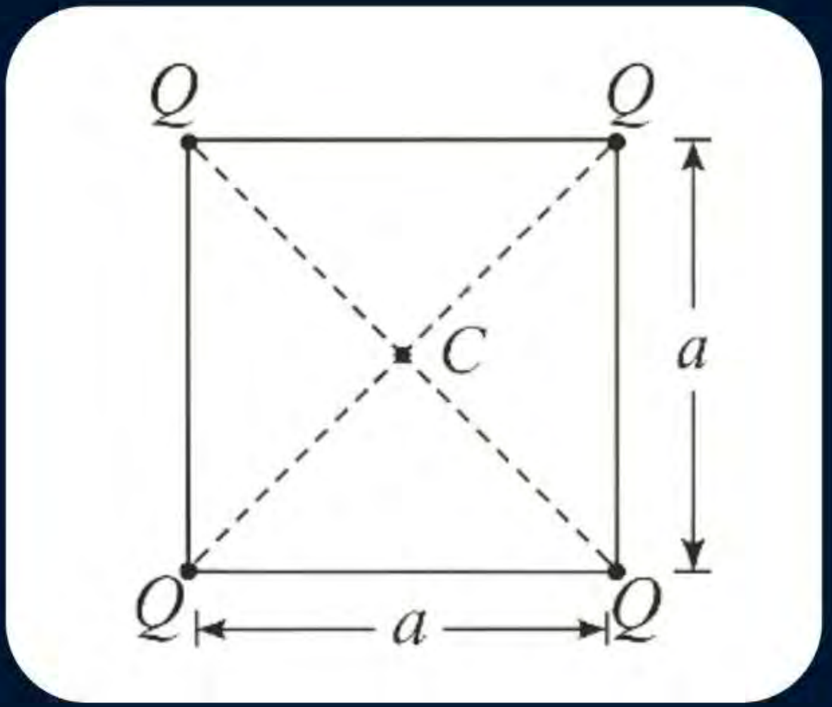
Potential due to System of Charges

HW



Potential due to System of Charges

HW



QUESTION

Note: Potential can be -ve.

Two unlike charges of magnitude q are separated by a distance 2d. The potential at a point midway between them is:



$V_P = ?$

$$V_P = \frac{kq}{d} + \frac{k(-q)}{d} = 0$$

A Zero

B $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{2d}$

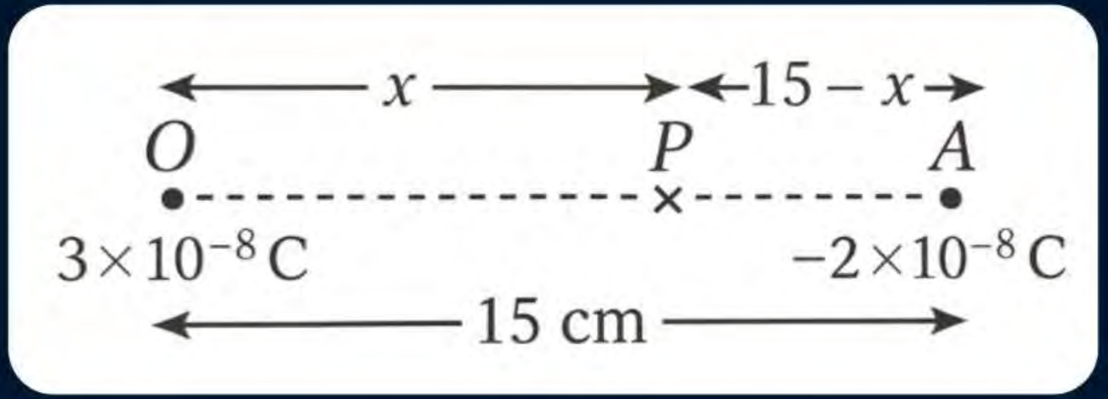
C $\frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{d}$

D $\frac{1}{4\pi\varepsilon_0} \cdot \frac{2q}{d}$



QUESTION

Two charges 3 x 10-8 C and -2 x 10-8 C are located 15 cm apart. At what point on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.



Potential due to an Electric Dipole

$$V_P = \frac{Kq}{r_1} + \frac{K(-q)}{r_2} = kq\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$P = 2aq$

$a \ll r$

$\Rightarrow a^2 \ll\ll r^2$

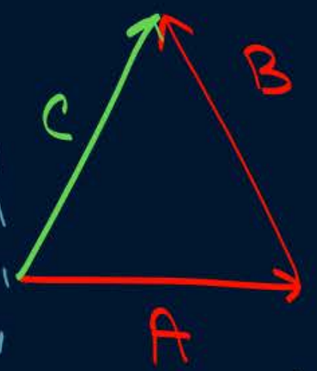
$\vec{a} + \vec{r_1} = \vec{r}$

$\Rightarrow \vec{r_1} = \vec{r} - \vec{a}$

$\Rightarrow \vec{r_1} = \vec{r} + (-\vec{a})$ —— (i)

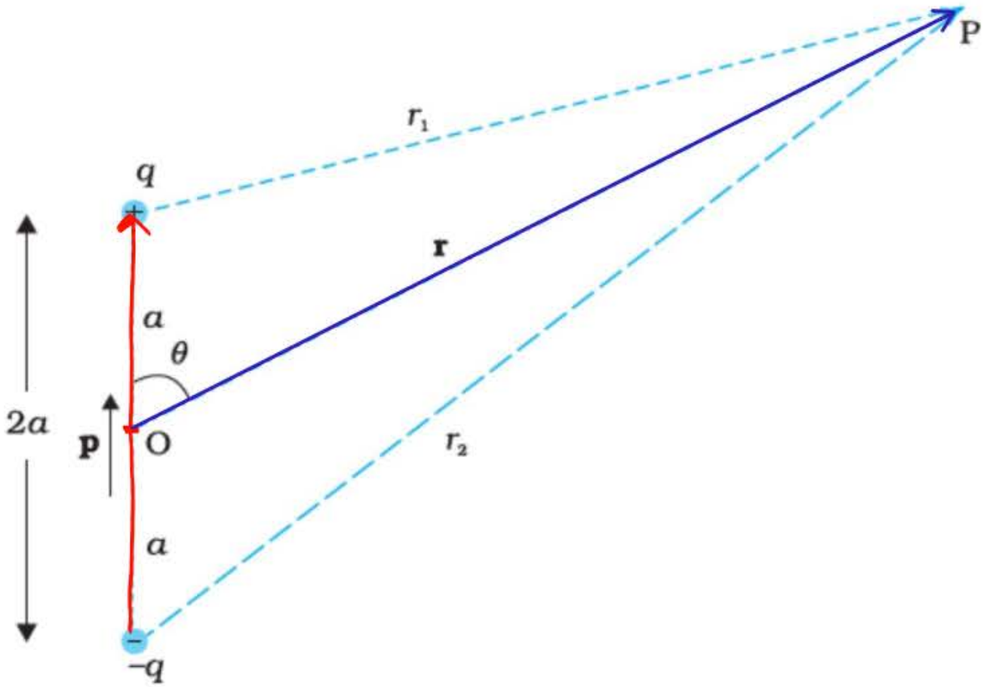
$\vec{a} + \vec{r} = \vec{r_2}$

$\Rightarrow \vec{r_2} = \vec{a} + \vec{r}$ —— (ii)



Triangle law of vector addⁿ.

$\vec{C} = \vec{A} + \vec{B}$



Quantities involved in the calculation of potential due to a dipole.

Magnitudes of $\vec{r}_1$ and $\vec{r}_2$:-

$$r_1 = \sqrt{r^2 + a^2 + 2(r)(-a)\cos\theta}$$

$$\Rightarrow r_1^2 = r^2 + a^2 - 2ra\cos\theta$$

$$\Rightarrow r_1^2 = r^2\left(1 - \frac{2a\cos\theta}{r} + \frac{a^2}{r^2}\right)$$



$$\Rightarrow r_1 = r\left(1 - \frac{2a\cos\theta}{r}\right)^{1/2}$$

$$\Rightarrow \frac{1}{r_1} = \frac{1}{r}\left(1 - \frac{2a\cos\theta}{r}\right)^{-1/2}$$

$$\Rightarrow \frac{1}{r_1} = \frac{1}{r}\left(1 + \frac{a\cos\theta}{r}\right) \quad \text{——— (iii)}$$

$$r_2 = \sqrt{r^2 + a^2 + 2(r)(a)\cos\theta}$$

$$\Rightarrow r_2^2 = r^2 + a^2 + 2ra\cos\theta$$

$$\Rightarrow r_2^2 = r^2\left(1 + \frac{2a\cos\theta}{r} + \frac{a^2}{r^2}\right)$$



$$\Rightarrow r_2 = r\left(1 + \frac{2a\cos\theta}{r}\right)^{1/2}$$

$$\Rightarrow \frac{1}{r_2} = \frac{1}{r}\left(1 + \frac{2a\cos\theta}{r}\right)^{-1/2}$$

$$\Rightarrow \frac{1}{r_2} = \frac{1}{r}\left(1 - \frac{a\cos\theta}{r}\right) \quad \text{——— (iv)}$$

Putting the final values of r_1 and r_2 in the expression of V :-

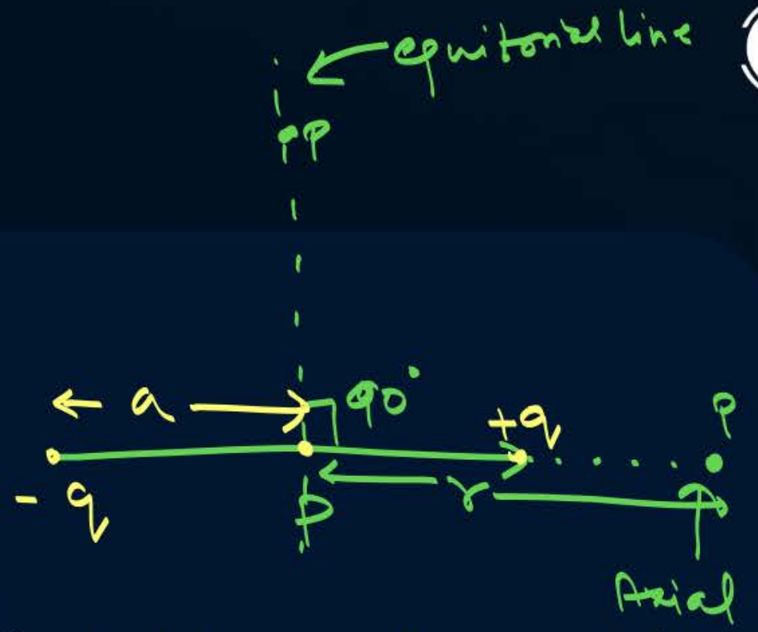
$$V_p = kq\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$$

$$\Rightarrow V_p = kq\left(\frac{1}{r}\left(1 + \frac{a\cos\theta}{r}\right) - \frac{1}{r}\left(1 - \frac{a\cos\theta}{r}\right)\right)$$

$$\Rightarrow V_p = \frac{kq}{r}\left[1 + \frac{a\cos\theta}{r} - 1 + \frac{a\cos\theta}{r}\right]$$

$$\Rightarrow V_p = \frac{kq\,2a\cos\theta}{r^2} \Rightarrow \boxed{V_p = \frac{k_p\cos\theta}{r^2}}$$

Potential due to an Electric Dipole



Special Cases:

(I) $\theta = 90°$] → Equitorial line

$$V_P = 0$$

(II) $\theta = 0°$] → Along the Dipole on the axial line.

$$V_P = \frac{kp}{r^2}$$

(III) $\theta = 180°$] → Opposite to the Dipole's direction on the axis.

$$V_P = -\frac{kp}{r^2}$$

(IV) At the midpoint of the Dipole.

$$V_P = 0$$

QUESTION

Find electric potential due to dipole at point P as showing in the figure-

$$V_P = \frac{kp\cos\theta}{r^2} = \frac{kp}{x^2}\frac{1}{2} = \frac{kp}{2x^2}$$

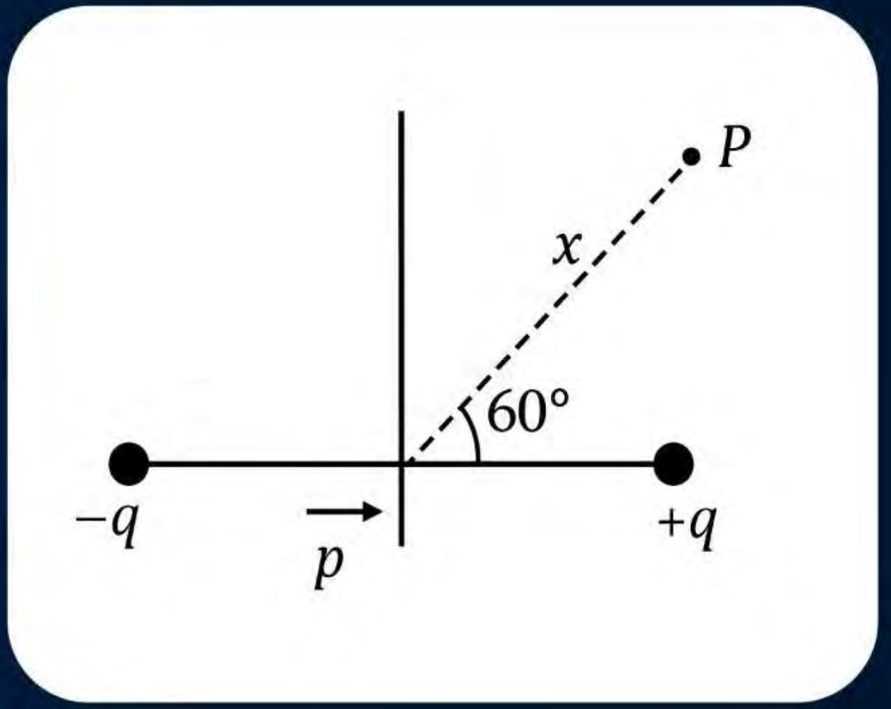
$$V_P = \frac{k(2aq)\cos\theta}{x}$$

A $\frac{kp}{2x^2}$

B $\frac{kp}{x^2}$

C $\frac{2kp}{x^2}$

D $\frac{kp}{3x^2}$



HW.

Basic Math

- Watch
- Make notes
- Ques Practise.

THANK YOU

